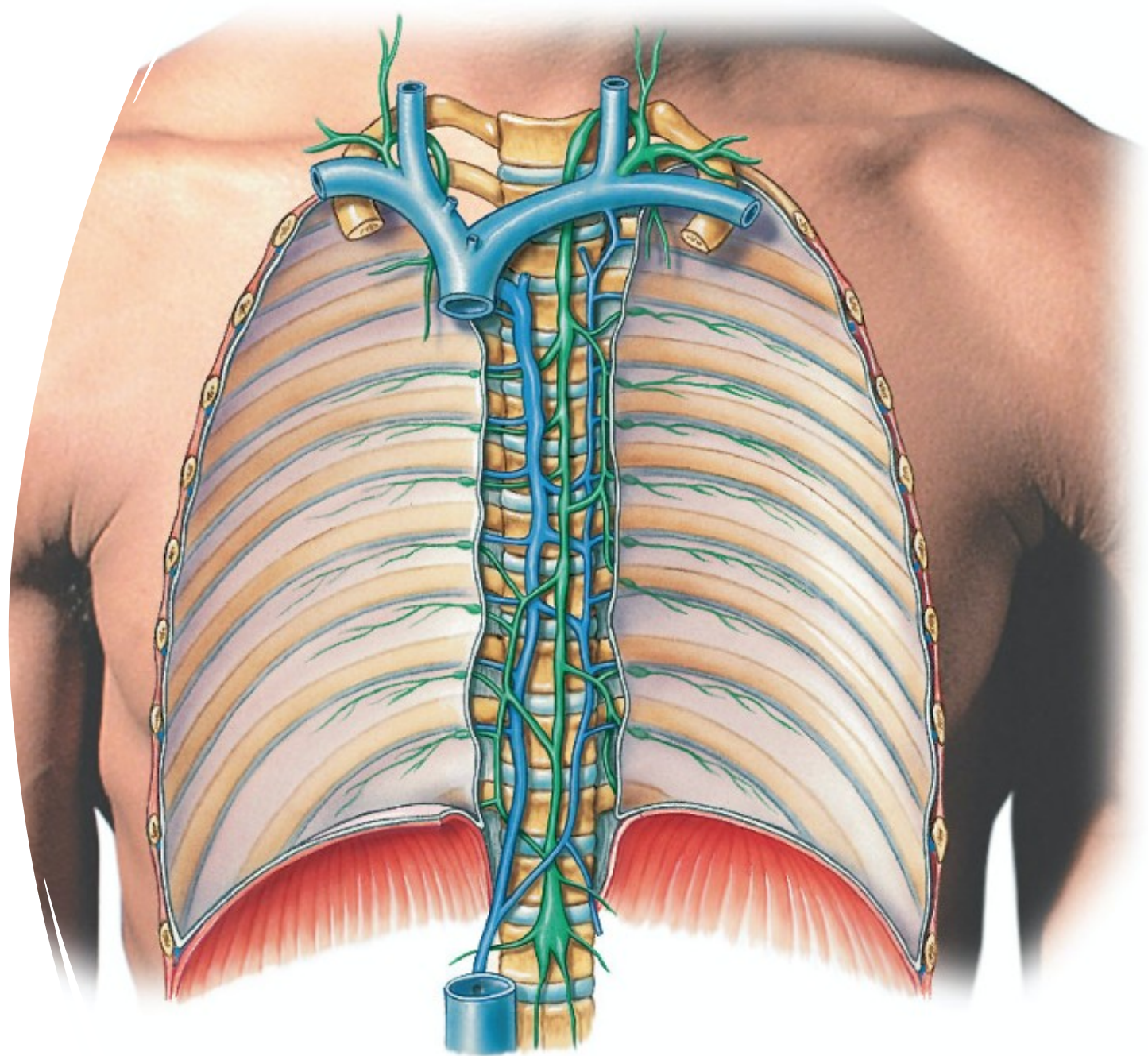


BIO262 – Physiology

Module 11
Lymph
Immune

What are the functions of the lymphatic system?

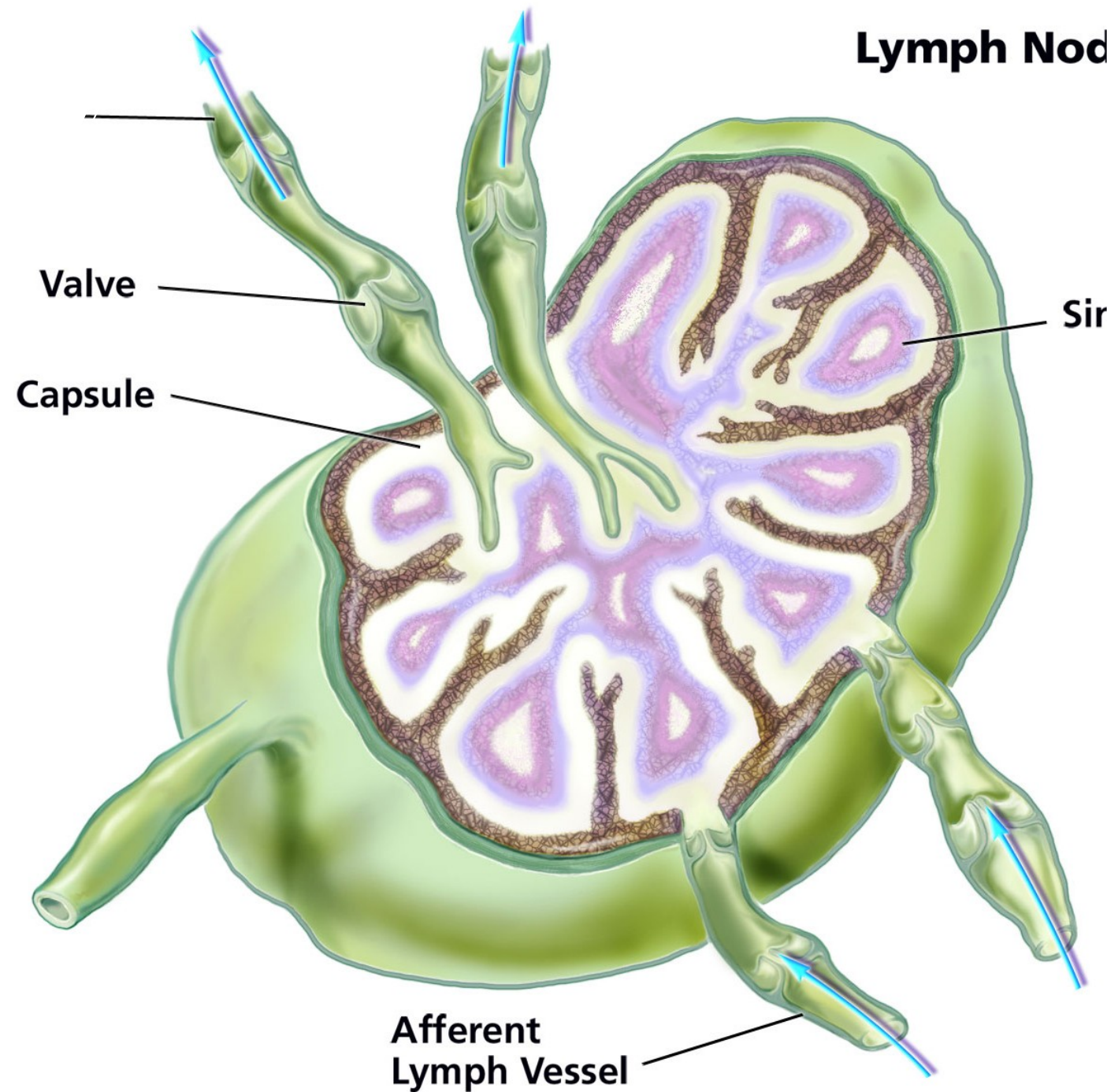
- Returns excess fluid from peripheral tissues to blood stream
- Lymph fluid:
 - Helps to transport immune cells throughout the system



What is the function of lymph nodes?

Function:

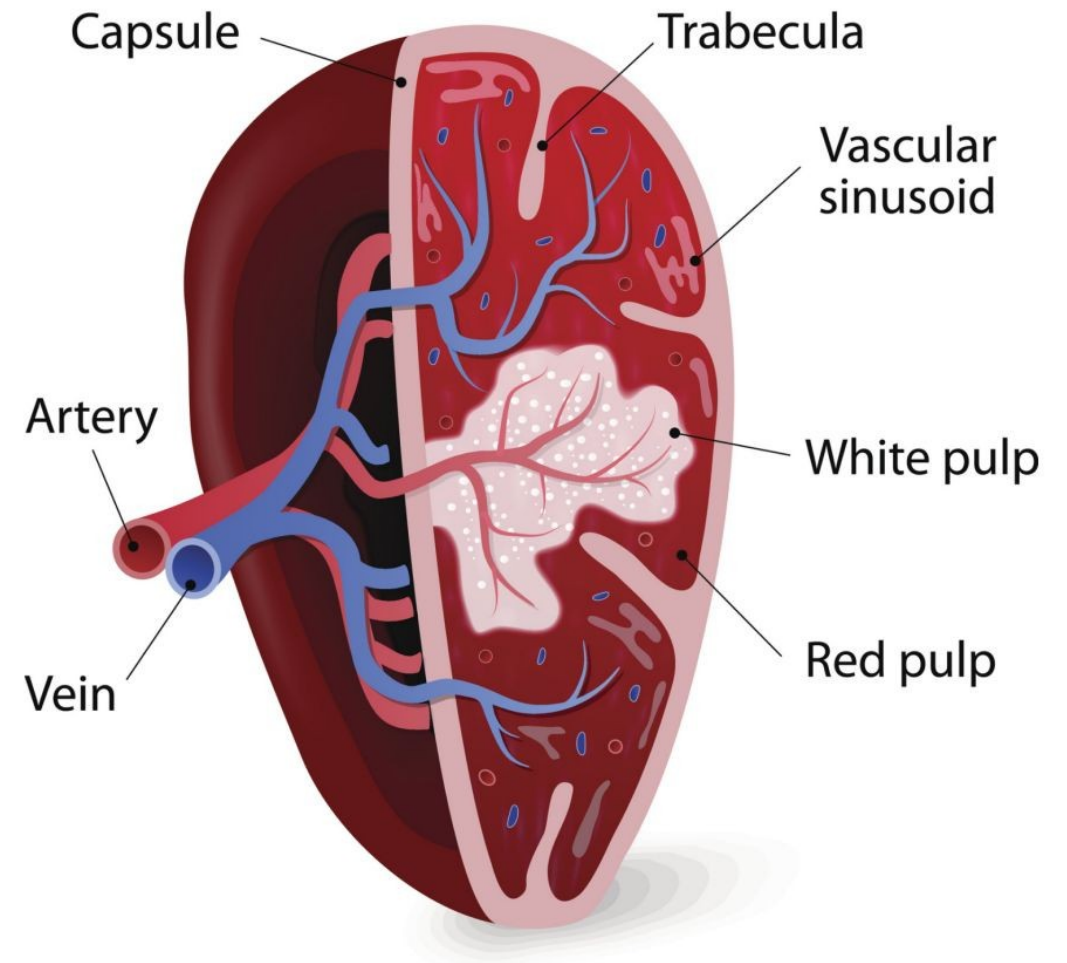
- Filters & purifies lymph before reaching venous circulation



Functions of the Spleen

1. Removes old red blood cells from circulation
2. Stores iron and platelets
3. Initiates immune response by B & T cells in response to circulating antigens

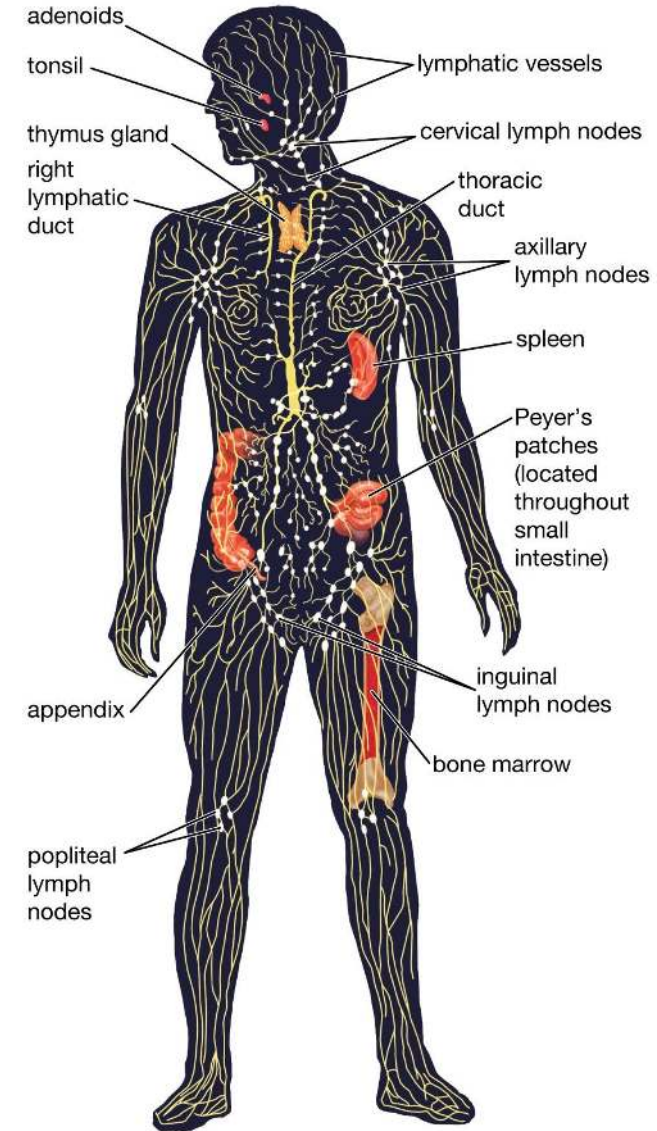
SPLEEN ANATOMY



Function of MALT

MALT : mucosa associated lymphoid tissue:

- Lymphoid tissue along mucosa of urinary, digestive & reproductive tracts that detect pathogens!
- Locations are significant in that they have openings to outside environment
 - **Peyer's Patches**: in epithelial lining of intestines
 - **Appendix**: found at cecum
 - **Tonsils**: found in pharynx



Lymphoid Organs: Matching

1. Lymph Nodes

2. Thymus

3. Spleen

A. Site for RBC death/removal

B. Is at its smallest relative size in the elderly

C. Contains red pulp and white pulp

D. Filters passing lymph to initiate immune responses to antigens

E. Becomes swollen when a local infection induces increased numbers of immune cells inside

F. Site of T cell maturation



The Immune System- 3 Lines of Defense

- Barriers
- Secretions

- Main cells: Macrophages and Neutrophils
- Non-specific, no memory, quick to respond

Innate Immunity (Non-specific Defense Mechanisms)		Adaptive Immunity (Specific Defense Mechanisms)
Timeline: 0 to 12 hours		Timeline: 1 to 7 Days
1 st Line of Defense	2 nd Line of Defense	3 rd Line of Defense
• Skin • Mucous membranes • Secretions of skin • Secretions of mucous membranes	• Macrophages • Other Phagocytes (i.e. neutrophils, NK cells) • Antimicrobial proteins • The Inflammatory response (e.g. redness, fever)	• Lymphocytes (B & T Cells) • Antigen-specific • Antibodies • Memory

The 1st Line of Defense includes the skin; mucous membranes; hair within the nose; cilia in the upper respiratory tract; urine; perspiration, saliva; stomach gastric juice; and sebum.

The 2nd Line of Defense includes an inflammatory response and white blood cells (called phagocytes) that ingest pathogens.

- Specific, has memory, slower to respond
- Lymphocytes (B and T cells)
- B cell - produce antibodies
 - T cell
 - T helper cells (CD4) - coordinate between B & T cells
- Cytotoxic T cell - directly approach the foreign substance & destroy.



What is innate immunity?



**FIRST LINE OF DEFENSE
“LET’S KEEP EM’ OUT”**



**DOES NOT DISTINGUISH BETWEEN
THREATS; IF IT IS FOREIGN, IT WILL BE
ATTACKED.**

3 Properties of Innate Immunity

Not Specific:

- Macrophages, Neutrophils and NK cells respond to any pathogen that enters the body. Same way every time!

Rapid:

- Due to not being specific it can jump right in

No Memory:

- Responds the same way every time.



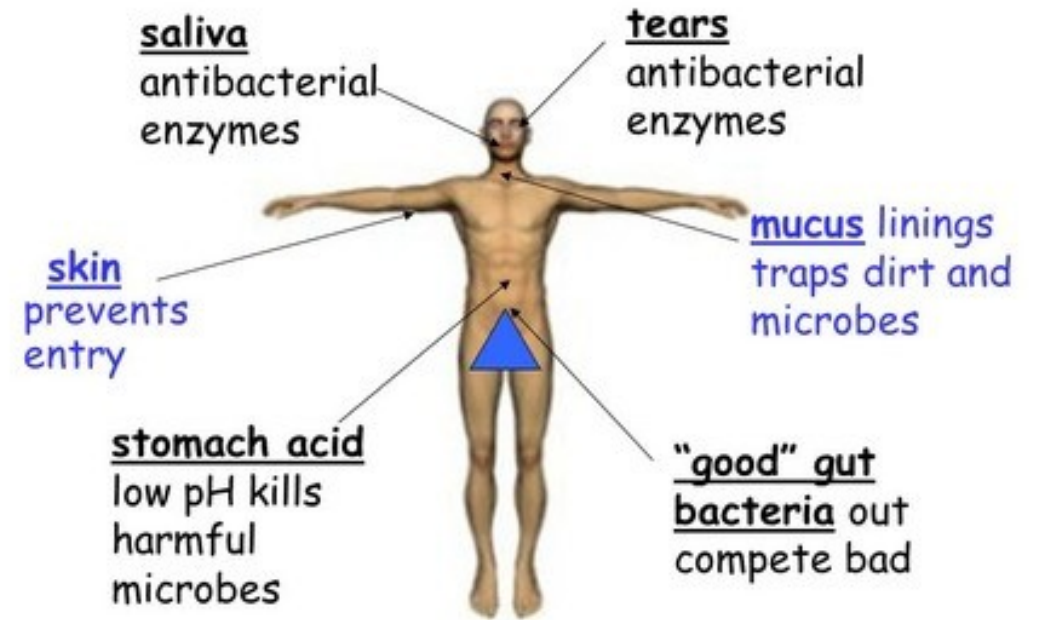
INNATE IMMUNITY

1. Prevent Entry:

- Skin, hair, nails
- Epithelial & mucosal linings
- Secretions of epithelium
 - sweat, sebum, mucous, urine, stomach acid



First lines of defense



INNATE IMMUNITY- Second Line of Defense

2. Phagocytes

*Neutrophils:

- Eat bacteria, debris or foreign compounds coated with antibodies

Eosinophils:

- Highly active in allergic reactions

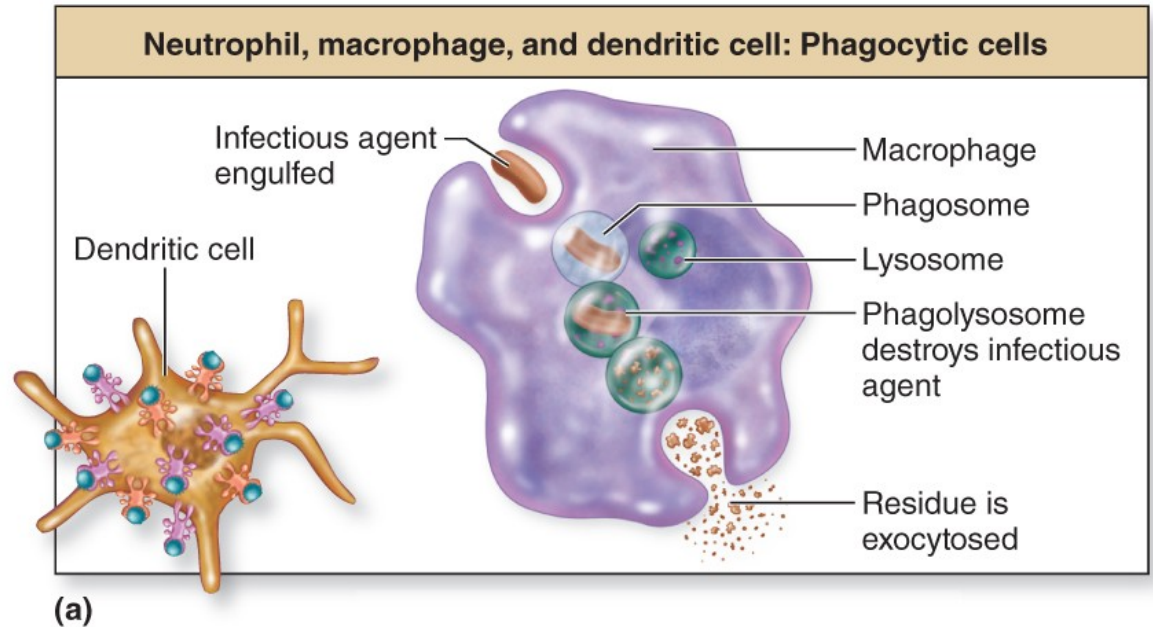
Dendritic cells:

- Antigen presenting cells, they capture antigens and present them to the T cells

*Macrophages:

- Engulf & destroy with lysosomal enzymes

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**What produces
all the WBC?**



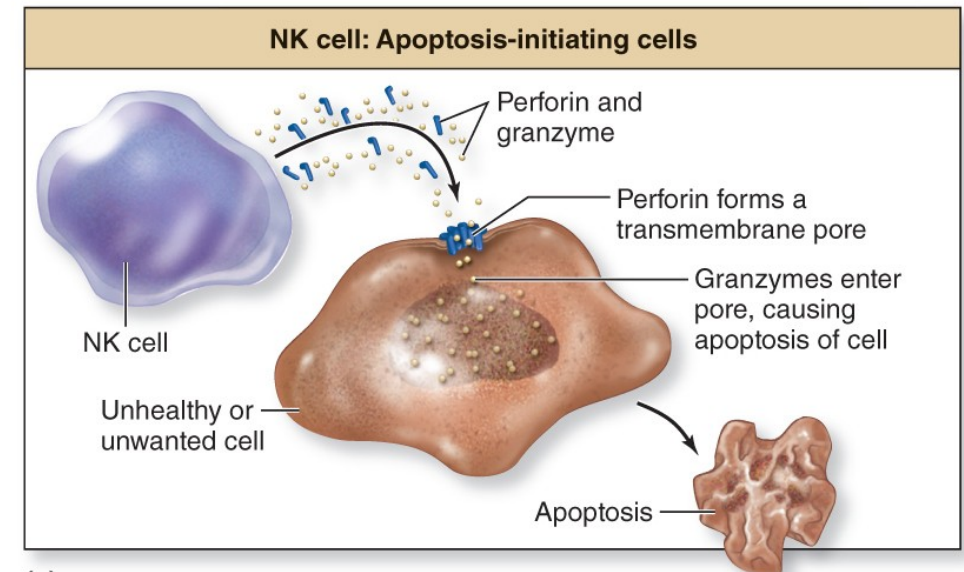
INNATE IMMUNITY

3. Immunological Surveillance

- Natural killer (NK) cells recognize & kill abnormal cells
 - Respond more quickly than B & T Cells & are less specific
 - Will attack any abnormal cell including bacteria, viruses, cancer cells etc.



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(c)

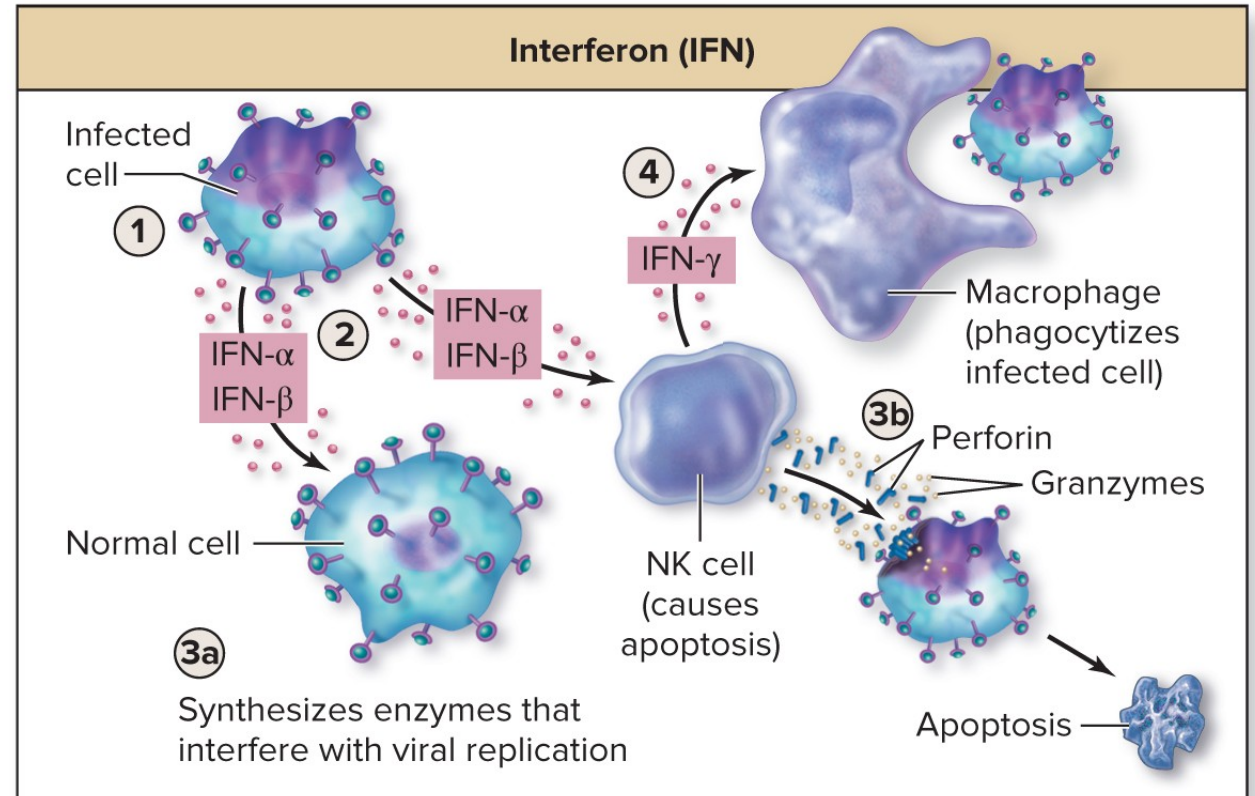


INNATE IMMUNITY

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4. Interferons

- chemicals released by cells
- coordinate defenses against viruses

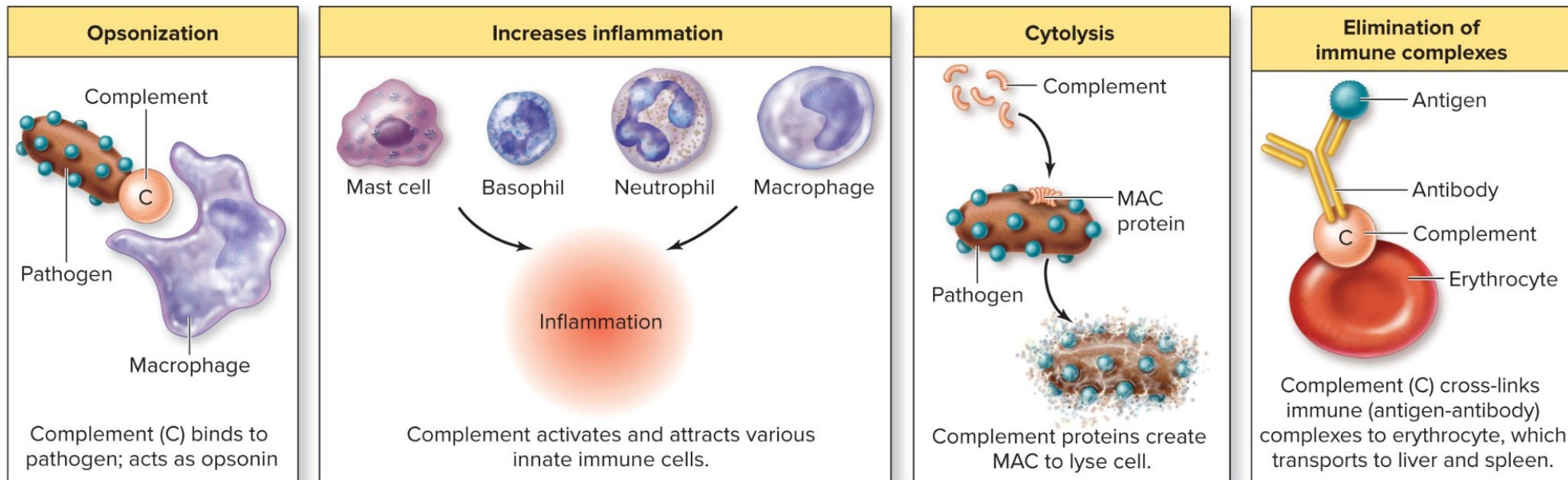


INNATE IMMUNITY

5. Complement system: 30+ proteins found in plasma

- Job is to tag the pathogen so that the phagocytic cells can find the pathogens and destroy them

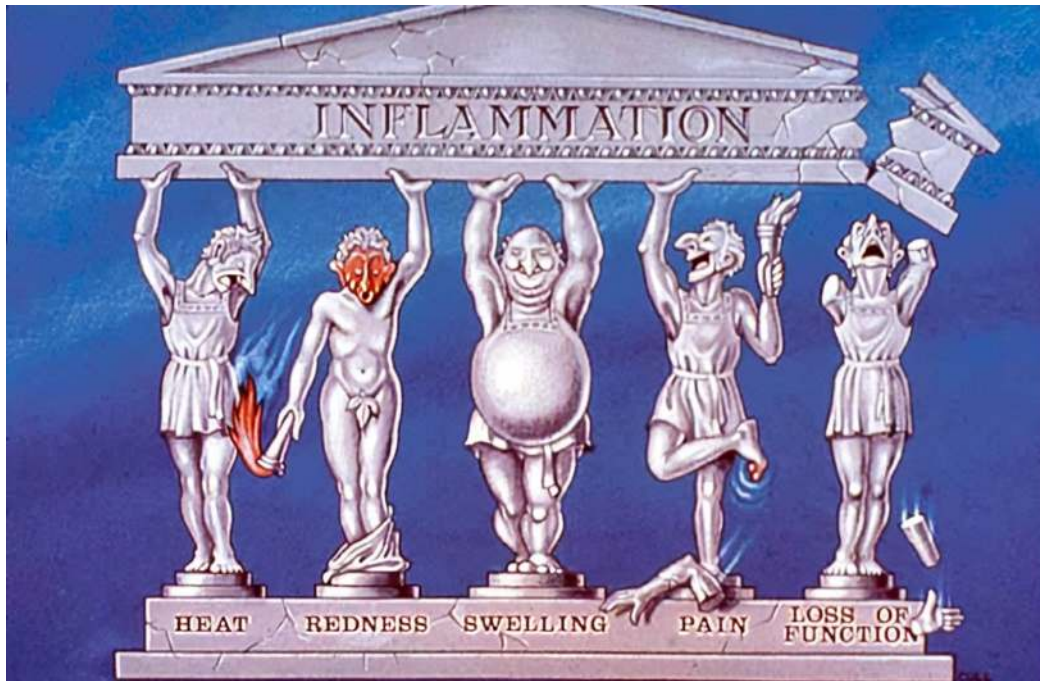
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INNATE IMMUNITY

6. Inflammatory Response:

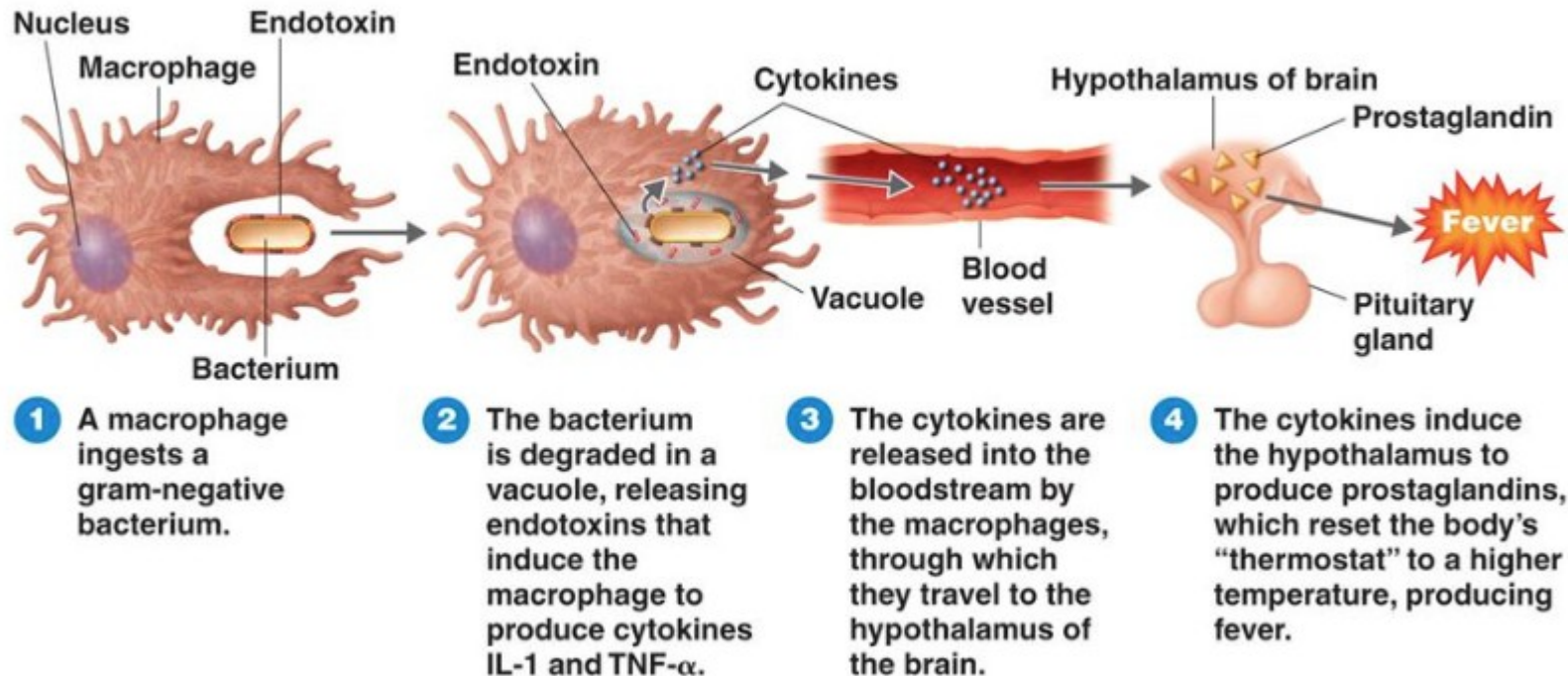
- limits spread of injury or infection



INNATE IMMUNITY

7. Fever: elevation of body temperature

- Accelerated metabolism to speed up defense processes
- Induced by circulating proteins that turn up the thermostat in the hypothalamus



4 Properties of Adaptive Immunity

1. Specificity:

- B cells, T-cells, and antibodies respond to a specific antigen.
- NOT NK CELLS (although they are a lymphocyte)

2. Versatility:

- many different lymphocytes

3. Memory:

- Recognizes the pathogen more rapidly the second time

4. Tolerance:

- Won't attack your own cells

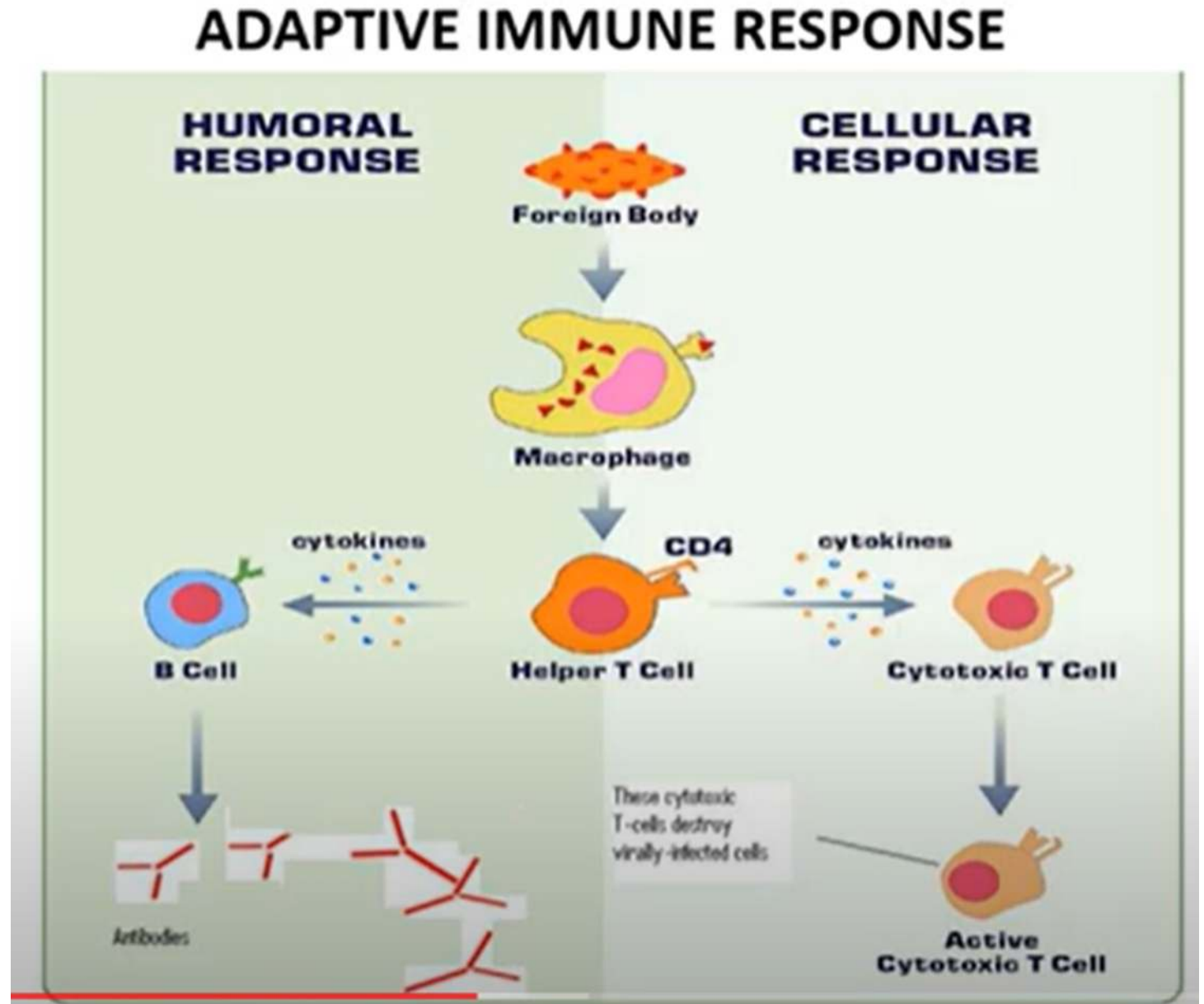
What cells are the main players?



Helper T-Cells

Helper T cells: CD4 Cells

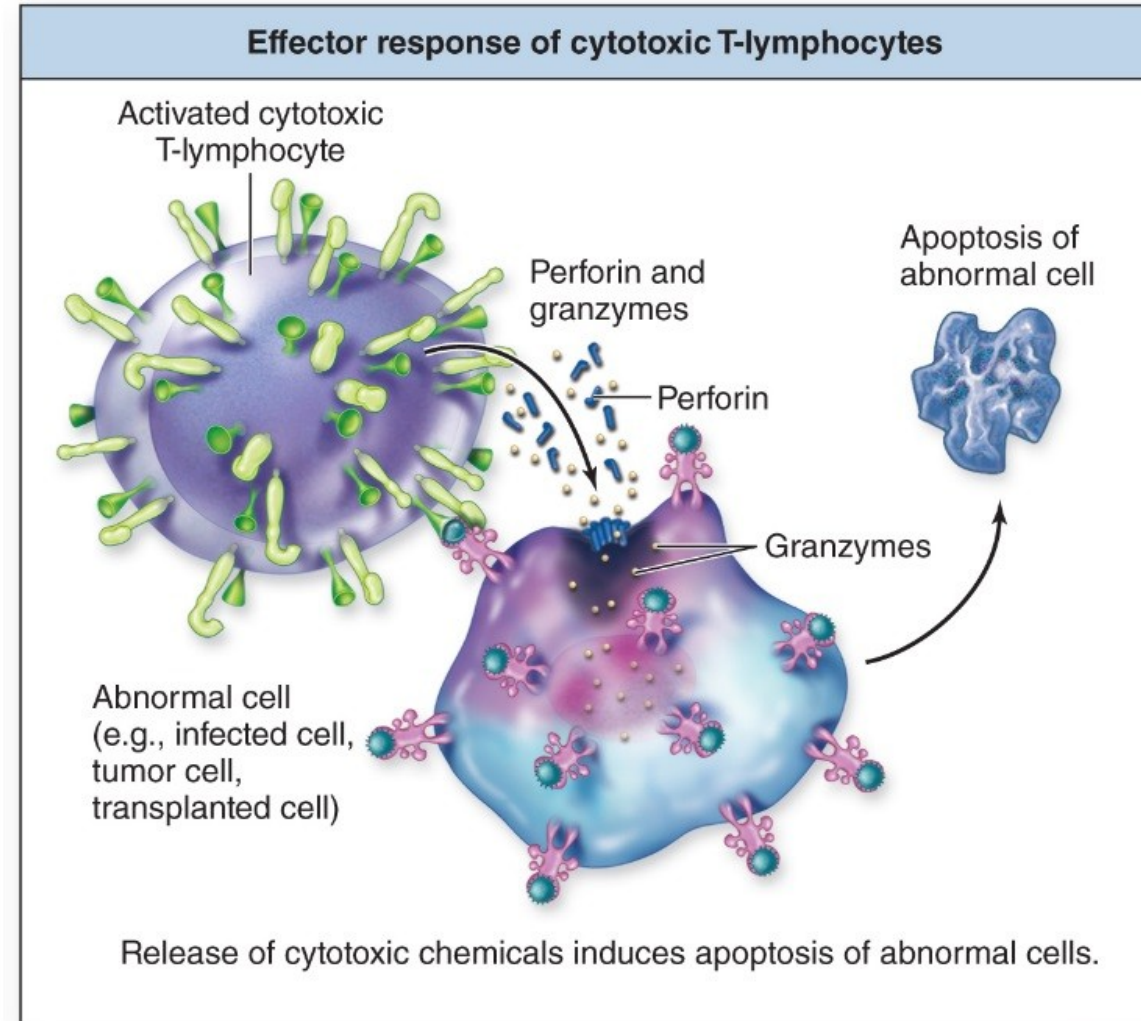
- Signals B cells and cytotoxic T cells and helps activate them
- Helps to coordinate the attack



Cytotoxic T-Cells

Cytotoxic T cells: CD8 cells

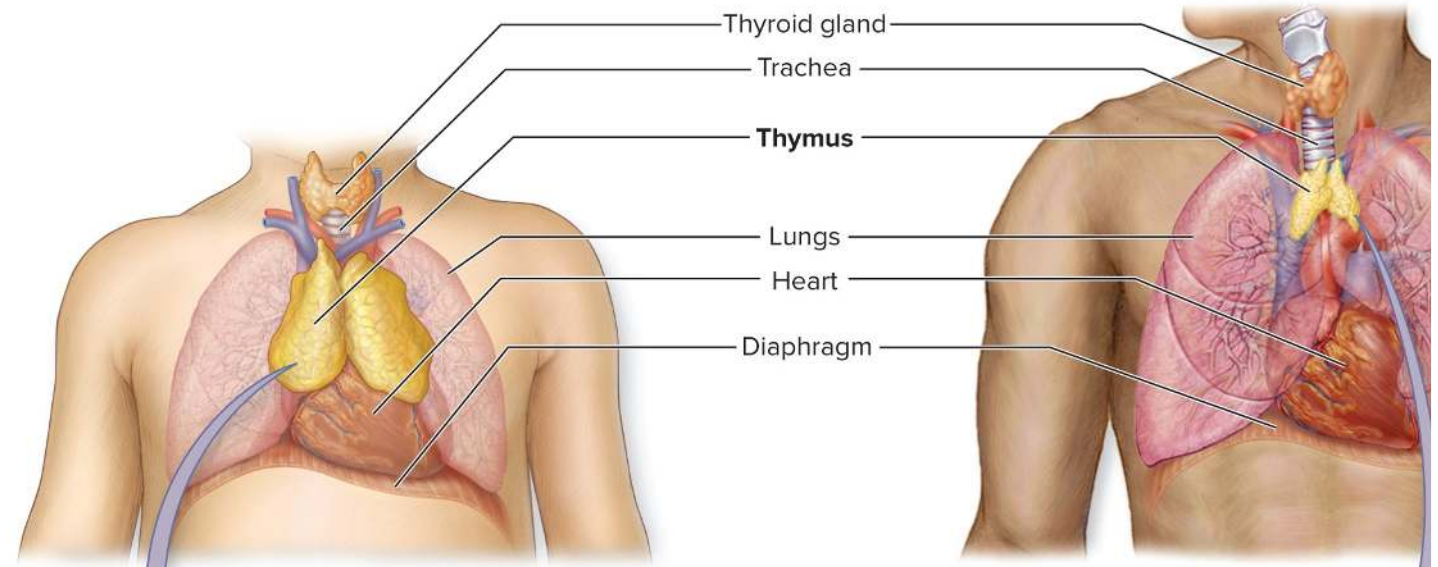
- Find & destroy abnormal cells that display its target antigen
- Like soldiers on the battlefield



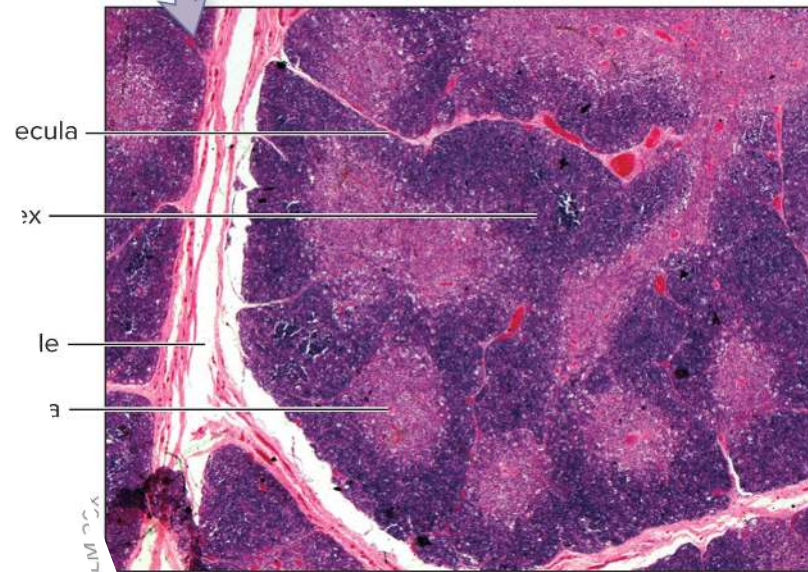
(b)

Function of the Thymus

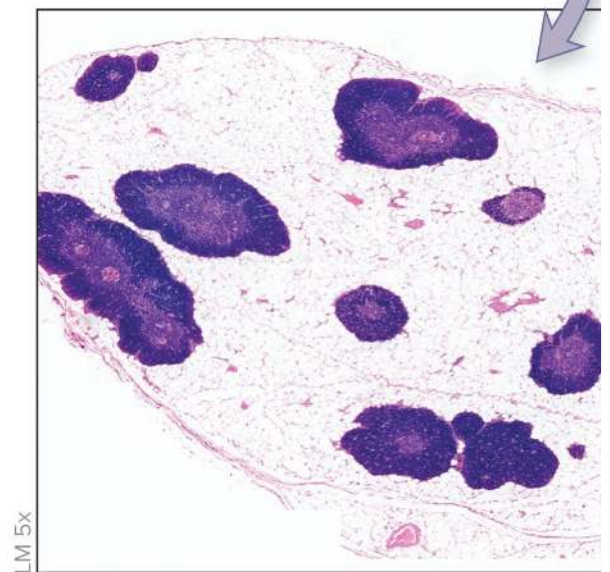
- Thymic hormones promotes maturation of T-cells
- Large in newborns and children
- Continues to decrease in size and function which may contribute to sickly elderly individuals



(a) Child (left) and adult (right) thorax, anterior view



(b) Micrograph of child's thymus

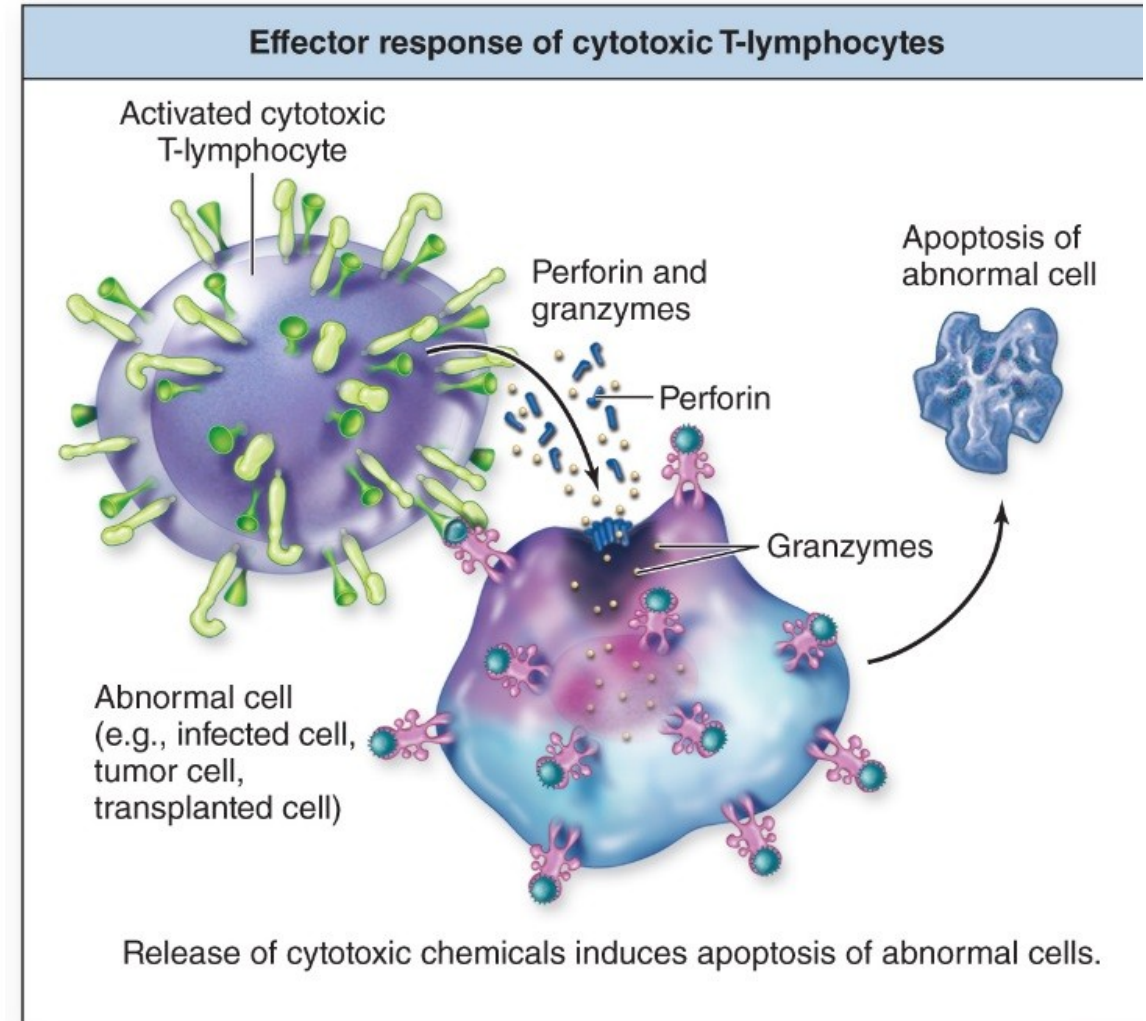


(c) Micrograph of adult's thymus

B-Cells

B cells:

- Make antibodies that are specific to one pathogen
- Like the factories making ammunition

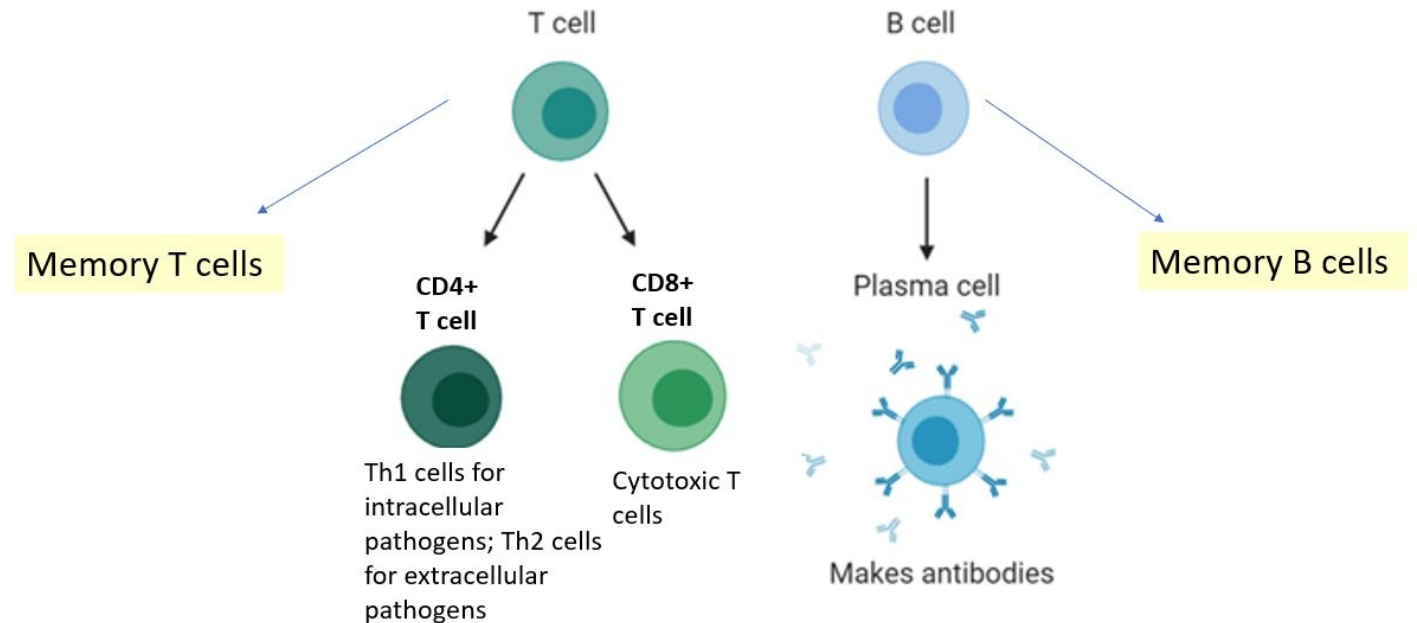


(b)

Memory cells

Memory cells:

- Can be T or B cells
- Long term immunity!
Can remember the specific infection next time you get it and mount a quicker response



EXAM PREP:

- Lymphocytes: know what they do and how they are activated. B cells, NK cells, Helper T cells, cytotoxic T cells, Memory cells, Plasma B cells. Note: only NK are part of innate immune response – the rest are part of adaptive.
- Lymph organs: what their job is/what happens in each – Thymus, spleen, MALT, lymph nodes.
- Lymph fluid – role and how it does this.
- Innate versus adaptive immunity: which is immediate, which takes time to get going. Role of complement. Which cells are present in which. Which one includes physical barrier (examples of what these are).



QUESTIONS?